

Design of a Bootstrapped Analog Sampling Switch in 180 nm CMOS

AIM:

To design and simulate a bootstrapped NMOS sampling switch using 180 nm CMOS technology and evaluate its suitability for high speed ADC front end applications by validating its performance against specified requirements.

Design Specifications:

Parameter	Value
Technology	180 nm CMOS
Supply Voltage	1.8 V
Input Signal Range	1 V _{pp} centered at 0.9 V
Maximum Input Frequency	≥ 10 MHz
Switch Type	Bootstrapped NMOS
On-Resistance Variation	$\leq \pm 5\%$
Minimum THD at 10 MHz	≤ -55 dB

Theory:

Problem with a Conventional NMOS Sampling Switch:

For a simple NMOS sampling switch operating in the linear region, the on-resistance R_{on} can be approximated as

$$R_{on} \propto \frac{1}{\mu_n C_{OX} (W/L) (V_{GS} - V_{TH})}$$

where

μ_n = electron mobility,

C_{OX} = oxide capacitance per unit area,

W/L = transistor aspect ratio,

V_{GS} = gate-to-source voltage,

V_{TH} = threshold voltage.

In a conventional NMOS switch, the gate is driven by a fixed clock voltage V_{CLK} , while the source follows the input signal V_{in} . Therefore,

$$V_{GS} = V_{CLK} - V_{in}$$

As V_{in} varies over its input range, V_{GS} also varies. This causes the on-resistance R_{on} to become signal-dependent, resulting in:

- variation of switch resistance with input amplitude,
- non-linear signal attenuation,
- harmonic distortion in the sampled output.

This behavior limits the performance of high-speed, high-resolution ADC front-end circuits.

Bootstrapped Switch Concept:

To overcome the limitation of signal-dependent on-resistance, a bootstrapped switch is employed. The key idea is to maintain a nearly constant gate-to-source voltage for the main sampling transistor, independent of the input signal level.

The bootstrapped circuit dynamically boosts the gate voltage such that

$$V_{GS} \approx V_{DD}$$

for the main sampling switch. Since V_{GS} remains approximately constant, the on-resistance also remains nearly constant over the entire input voltage range, thereby reducing distortion and improving linearity.

Working:

The operation of the bootstrapped sampling switch can be explained using two clock-controlled phases:

- Hold phase (CK = 0)
- Sampling / Tracking phase (CK = 1)

1. Hold Phase (CK = 0)

When the clock signal is low, the circuit prepares the bootstrap capacitor for the next sampling operation.

Operation during hold phase

1. Main switch OFF

The gate of the main sampling transistor is pulled close to ground through the clock-controlled network. Hence,

$$V_{GS} \approx 0$$

and the input signal is disconnected from the output.

2. Output hold

The load capacitor C_1 holds the previously sampled voltage. This corresponds to the hold operation of the sample-and-hold circuit.

3. Bootstrap capacitor charging

During this phase, the bootstrap capacitor C_b is connected between V_{DD} and ground (or an internal reference) through auxiliary transistors. As a result,

$$V_{Cb} \approx V_{DD}$$

This stored charge will be used to boost the gate voltage in the next phase.

2. Sampling / Tracking Phase (CK = 1)

When the clock goes high, the stored charge on the bootstrap capacitor is used to turn ON the main sampling switch with a boosted gate voltage.

Operation during sampling phase

1. Bootstrap capacitor reconfiguration

The clock-controlled transistors disconnect C_b from V_{DD} and reconnect it such that:

- One terminal of C_b is connected to the input voltage V_{in} ,
- the other terminal is connected to the gate of the main sampling transistor.

Since the capacitor was previously charged to V_{DD} ,

$$V_{Cb}=V_{DD}$$

2. Gate voltage boosting

Because one terminal of the capacitor follows V_{in} , the gate voltage rises by the same stored voltage:

$$V_G=V_{in}+V_{DD}$$

This process is referred to as **bootstrapping**.

3. Constant gate-source voltage

The source is approximately at V_{in} . Therefore,

$$V_{GS}=V_G-V_{in}$$

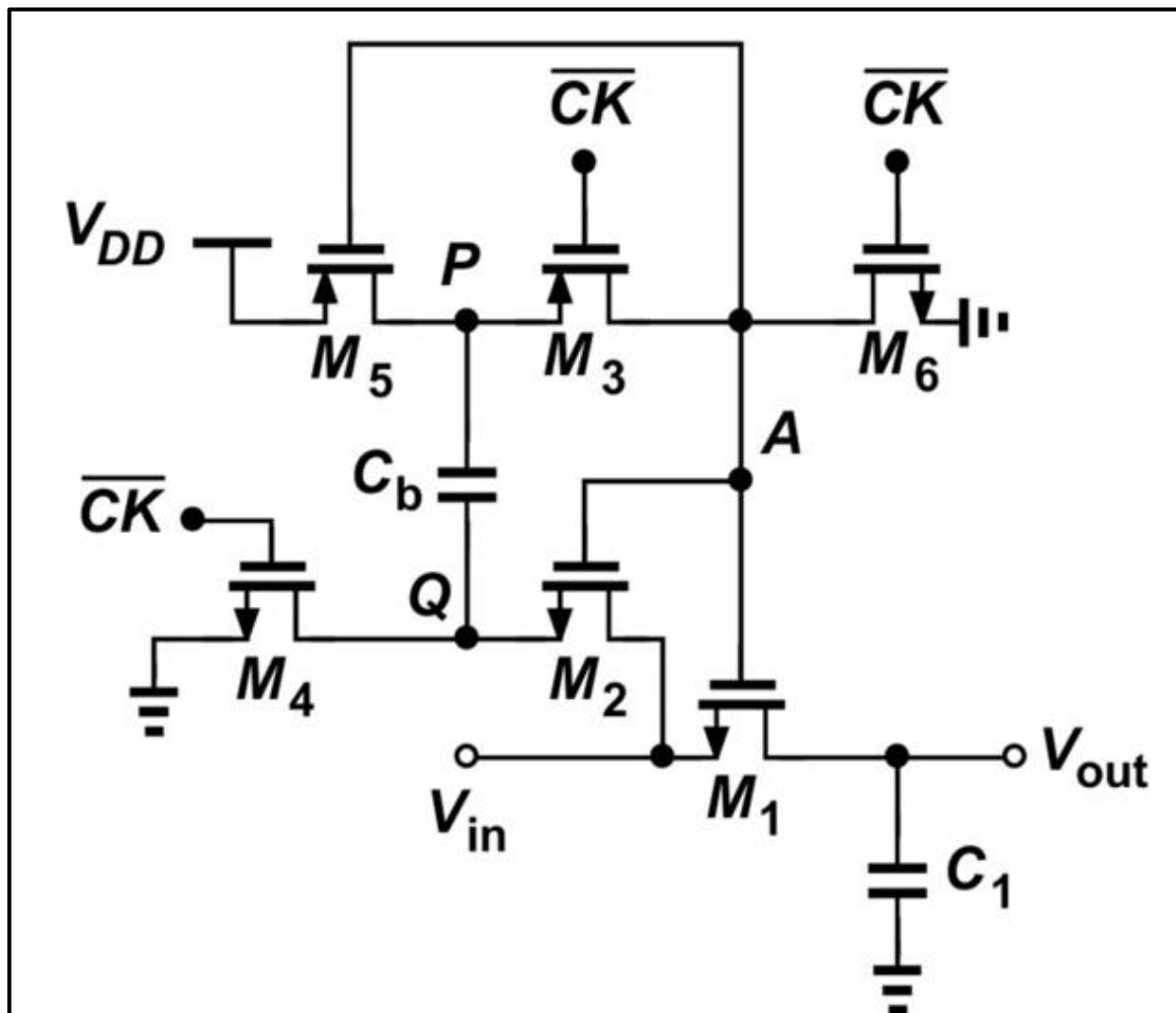
Substituting for V_G ,

$$V_{GS}=(V_{in}+V_{DD})-V_{in}=V_{DD}$$

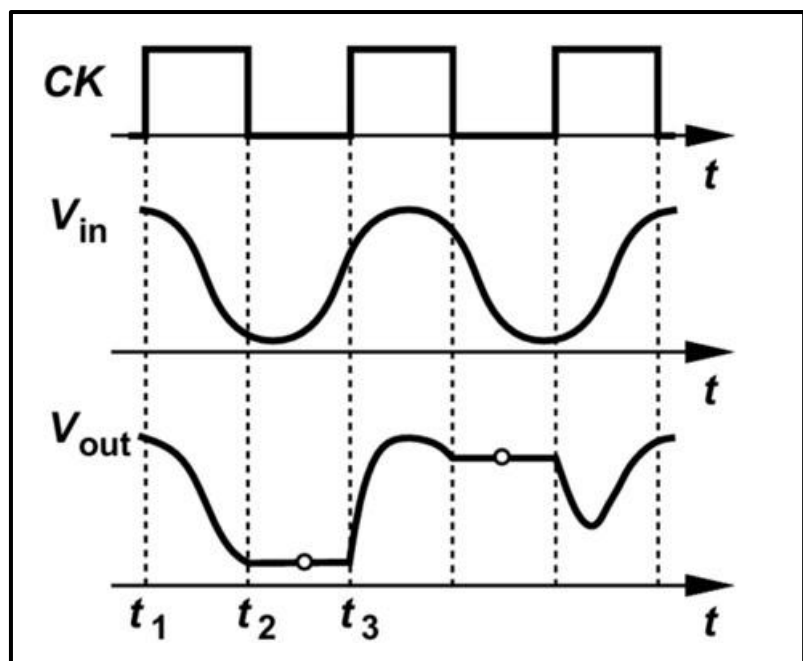
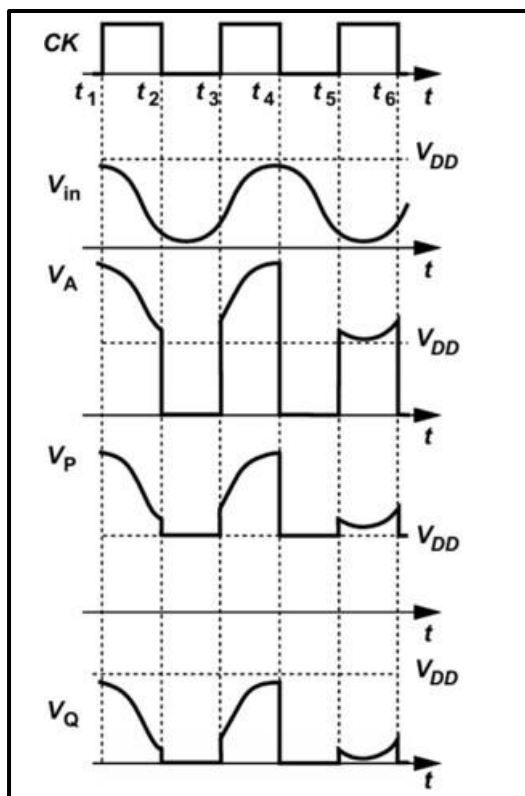
$$V_{GS}\approx V_{DD}$$

Hence, the gate-to-source voltage remains approximately constant and independent of the input signal level.

Circuit Diagram:



Waveform:



1. Output and Expected Waveforms

The screenshot displays the Virtuoso Schematic Editor interface. The main window shows a schematic diagram for a component named "ASHIT_RAJ_23BVD0025". The diagram is a complex circuit with multiple stages, including resistors (R), capacitors (C), and logic gates (A, Q). The components are interconnected with various signals and power supplies (Vdd, Vss, Ck, Ckb, Vin, Vout). The interface includes a Navigator on the left, a Property Editor at the bottom left, and a Command Line at the bottom. The title bar indicates the file name and the date/time: "Virtuoso® Schematic Editor L Editing: 23bvd0025_23bvd0025 schematic * (on cadence-server.vit.ac.in) Feb 20 17:38".

The screenshot displays the Cadence Virtuoso Analog Design Environment (ADE) interface. The main window shows a schematic of a circuit titled "23bvd0025_umc Bootstrap_SW_TB.23BVD0025 schematic * (on cadence-server.vit.ac.in)". The circuit is a boot strap switch, featuring a central block labeled "Boot Strap_Switch". This block has four inputs: "Ck", "Ckb", "Vin", and "Vss", and four outputs: "A", "P", "Q", and "Vout". The schematic is populated with various components, including capacitors (Ck, Ckb), voltage sources (V1, V2, V0, V3, V4), and a switch (V1=1.8, V2=1.8, tr=100p). The circuit is connected to a power supply network (Vdd, Vss) and a ground (gnd). The interface includes a top menu bar (Launch, File, Edit, View, Create, Check, Options, Window, PWS, UMC Utils, Help), a Navigator panel on the left showing the hierarchy of the design, and a Property Editor at the bottom left. The status bar at the bottom indicates the current file is "L: schSingleSelectPID" and the simulator is "spectre_aps".

Output Waveform Simulation

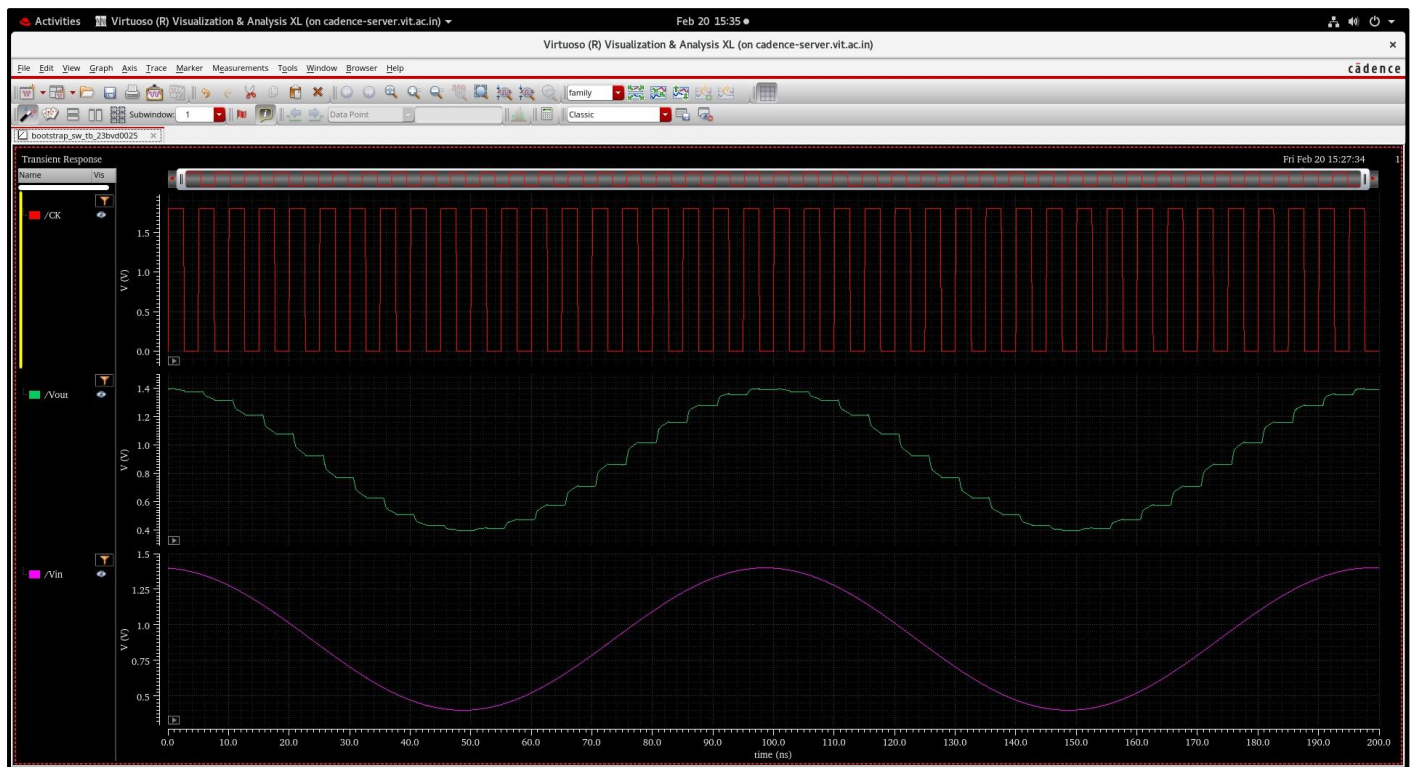
Transient simulation results verify the correct operation of the bootstrapped sampling switch.

- When the clock signal C_K is **low**, the sampling switch is turned OFF and the output voltage remains constant, indicating a proper **hold operation**.
- When the clock signal C_K is **high**, the sampling switch turns ON and the output voltage closely follows the input signal (V_{in}), confirming correct **sampling/tracking operation**.

These observations clearly demonstrate that the sample-and-hold functionality of the circuit is working as intended.

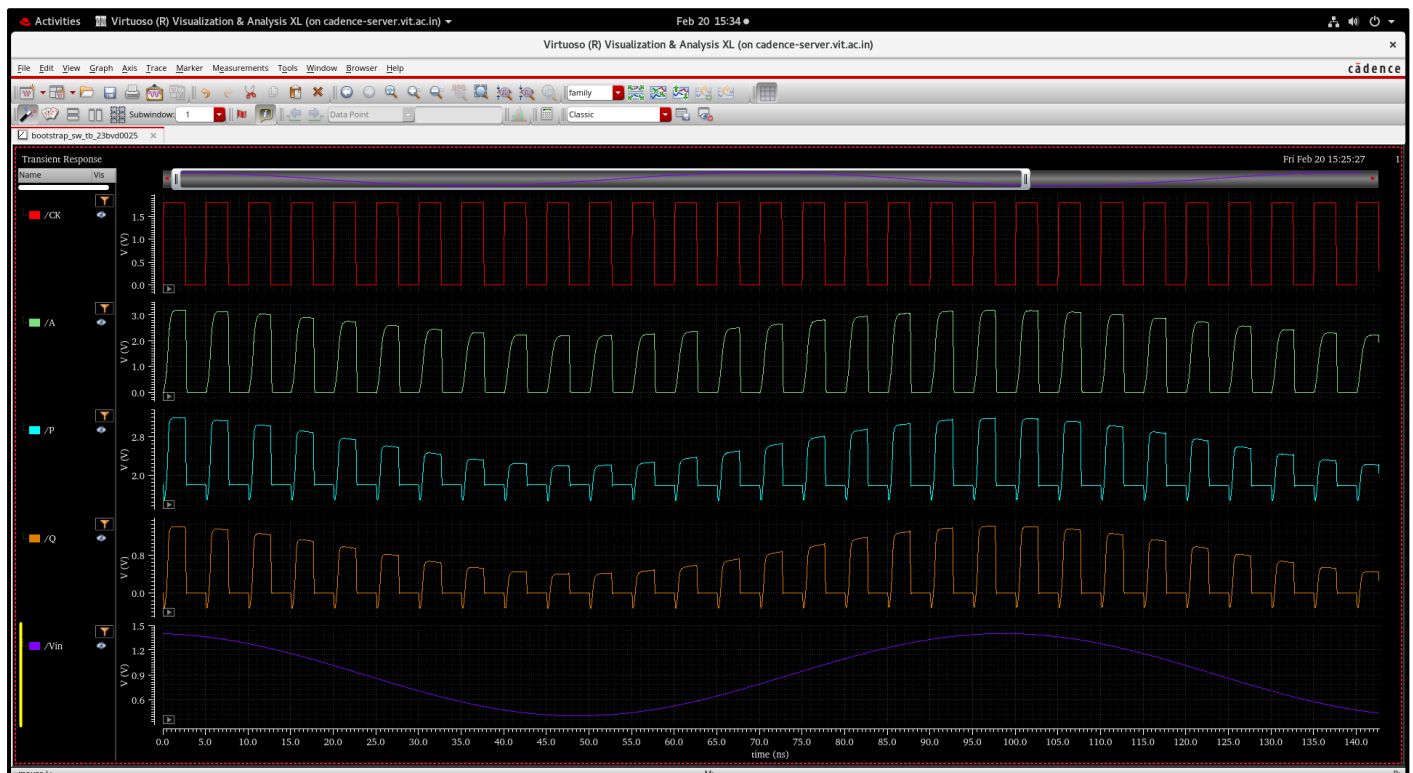
Sample-and-Hold Operation

The sample-and-hold behavior is evident from the waveform plots, where the output tracks the input during the sampling phase and retains the sampled value during the hold phase.



Other Node Waveforms

Waveforms of internal nodes confirm correct biasing and signal levels. It is clearly observed that the input signal has a peak-to-peak amplitude of **1 V_{pp}**, centered at **0.9 V**, as required by the design specifications.



Conclusion (I)

The bootstrapped sampling switch operates exactly as expected:

- The output voltage is held constant when **CK = 0**
- The output follows the input signal when **CK = 1**

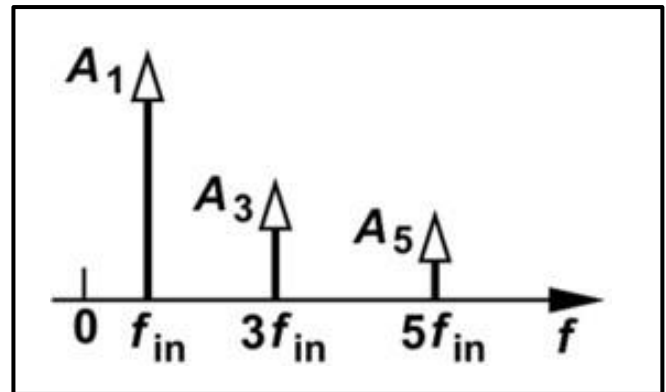
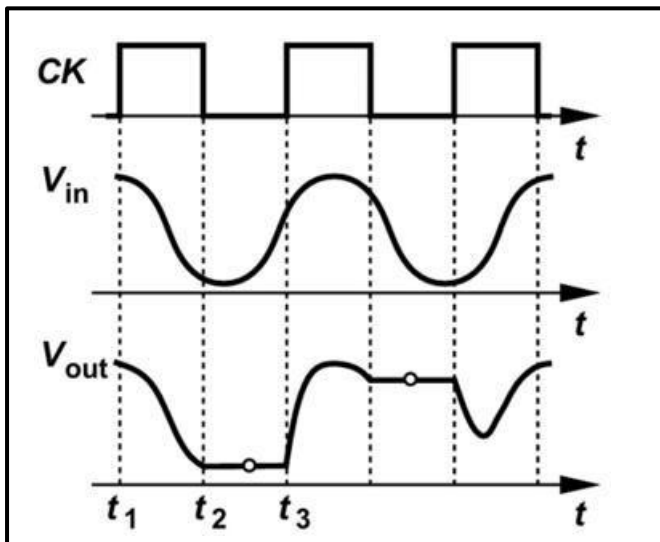
This confirms correct transient behavior and validates the basic functionality of the design.

2. Total Harmonic Distortion (THD) Analysis

Definition of THD

Total Harmonic Distortion (THD) quantifies the nonlinearity introduced by the sampling switch. In MOS-based sampling circuits, distortion primarily arises from the nonlinear dependence of the switch on-resistance on the input signal voltage.

As the effective gate-to-source voltage varies, the on-resistance becomes nonlinear, generating higher-order harmonics when a sinusoidal input is applied. Additional contributors to distortion include incomplete settling, charge injection, and clock feedthrough.

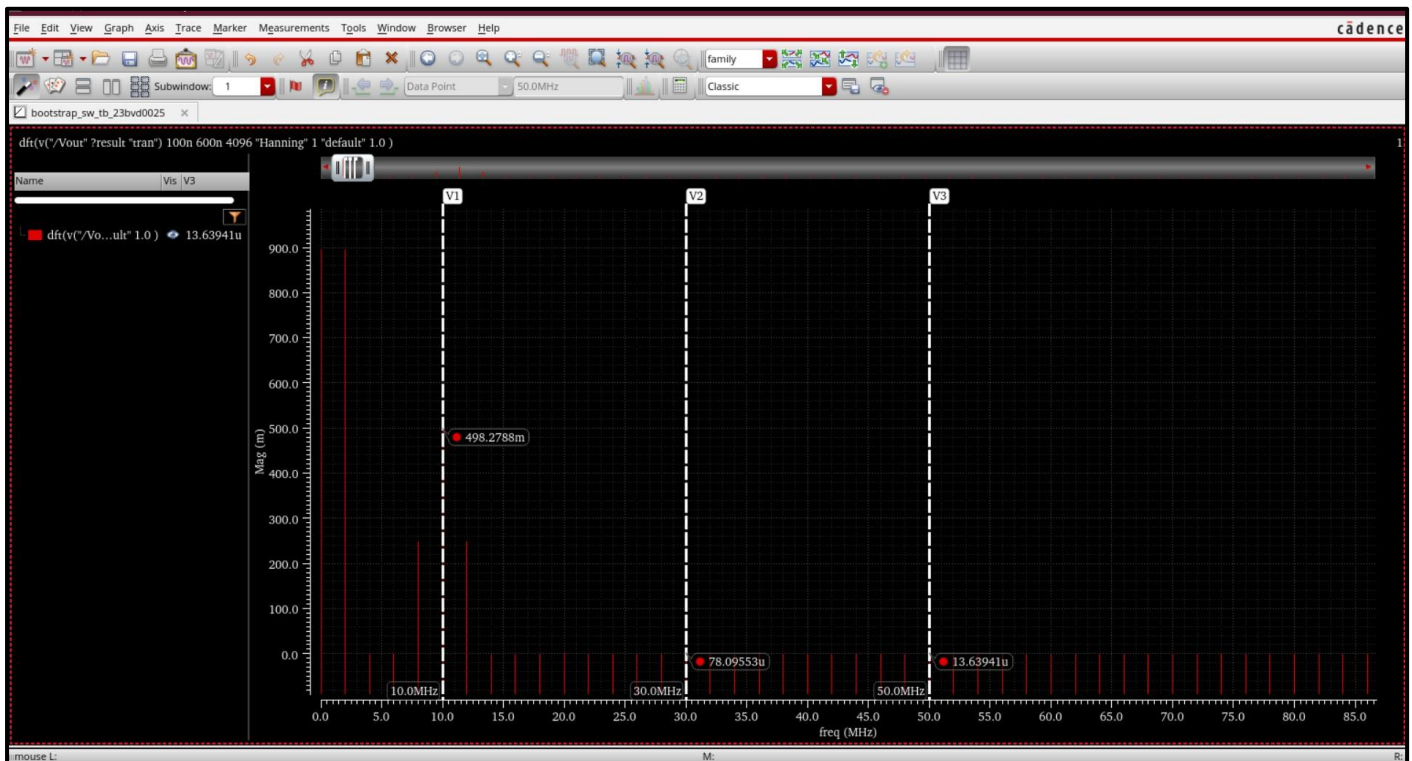


$$HD_{tot} = \frac{A_3^2 + A_5^2 + \dots}{A_1^2}.$$

FFT-Based THD Calculation

The harmonic content of the sampled output signal was evaluated by performing an FFT analysis. The measured spectral components are:

$$THD_{tot} = \frac{A_3^2 + A_5^2 + \dots}{A_1^2}$$



Substituting the measured harmonic amplitudes:

- Fundamental amplitude : $A_1 = 498.2788\text{mV} = 0.4982788\text{V}$
- Third harmonic amplitude : $A_3 = 78.09553\text{uV} = 78.09553 \times 10^{-6}\text{V}$
- Fifth harmonic amplitude : $A_5 = 13.63941\text{uV} = 13.63941 \times 10^{-6}\text{V}$

$$\text{THD}_{\text{tot}} = \frac{(78.09553 \times 10^{-6})^2 + (13.63941 \times 10^{-6})^2}{(0.4982788)^2} = -2.53138 \times 10^{-8}$$

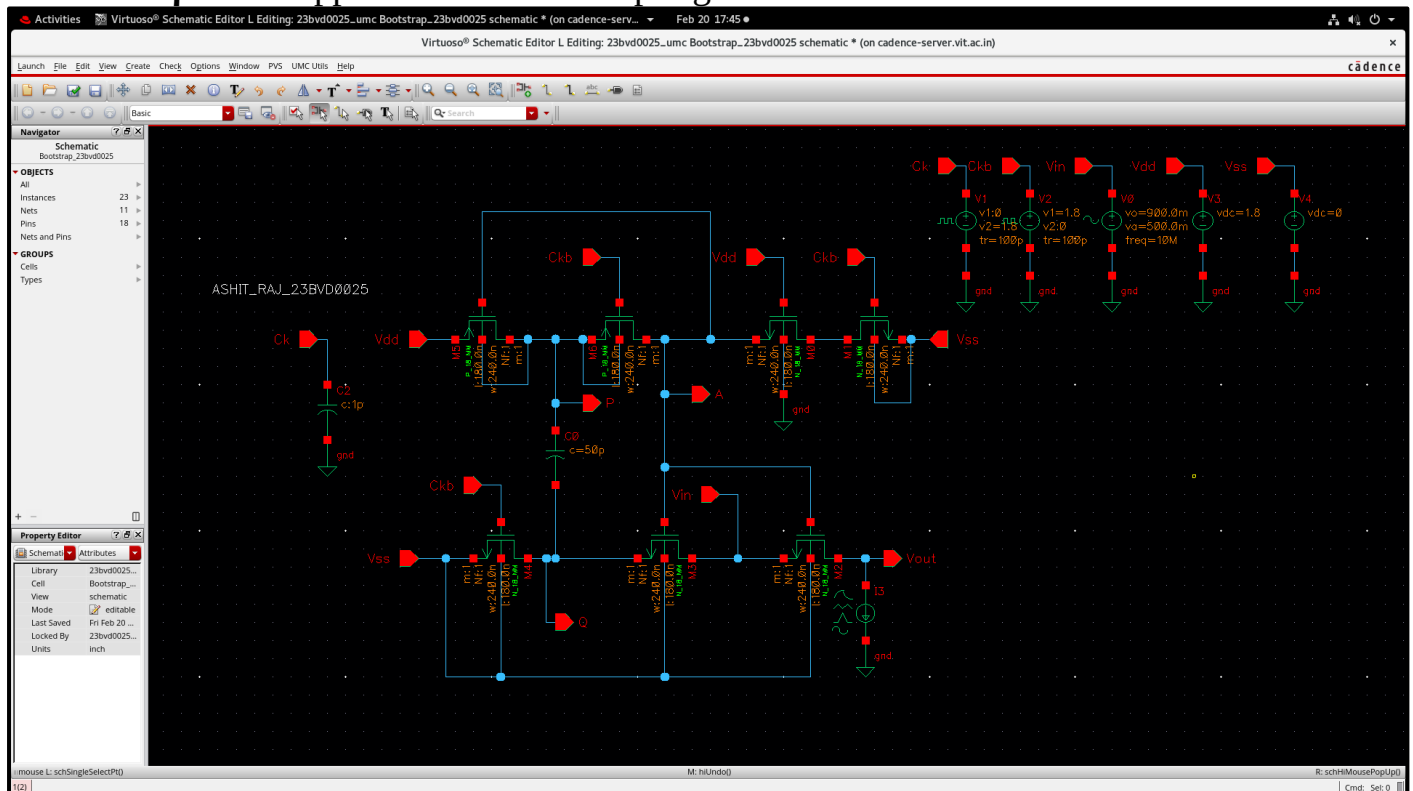
$$\text{THD (dB)} = 10 \log_{10}(-2.53138 \times 10^{-8}) = -75.99643 \text{ dB}$$

Conclusion (II)

The obtained THD value of **-75.99 dB** is significantly lower than the required limit of **-55 dB**, thereby satisfying the linearity requirement of the design.

3. On-Resistance (R_{ON}) Variation Analysis

To evaluate the variation of on-resistance across the input voltage range, a **constant current load of 10 μA** was applied across the sampling switch.



This current value was chosen to:

- produce a small voltage drop (≈ 0.5 mV),
- ensure operation in the linear (triode) region,
- avoid disturbance to normal circuit operation.

$$R_{avg} = \left(\frac{R_{MAX} + R_{MIN}}{2} \right)$$

$$\% variation = \left(\frac{R_{MAX} - R_{MIN}}{R_{avg}} \right) \times 100 \%$$

Measured On-Resistance Values

- At maximum input voltage:

$$R_{on,max} = 31.86 \, \Omega$$

- At minimum input voltage:

$$R_{on,min} = 28.54 \, \Omega$$

Step 1: Average Resistance

From simulation:

- $R_{max} = 31.86 \, \Omega$
- $R_{min} = 28.54 \, \Omega$

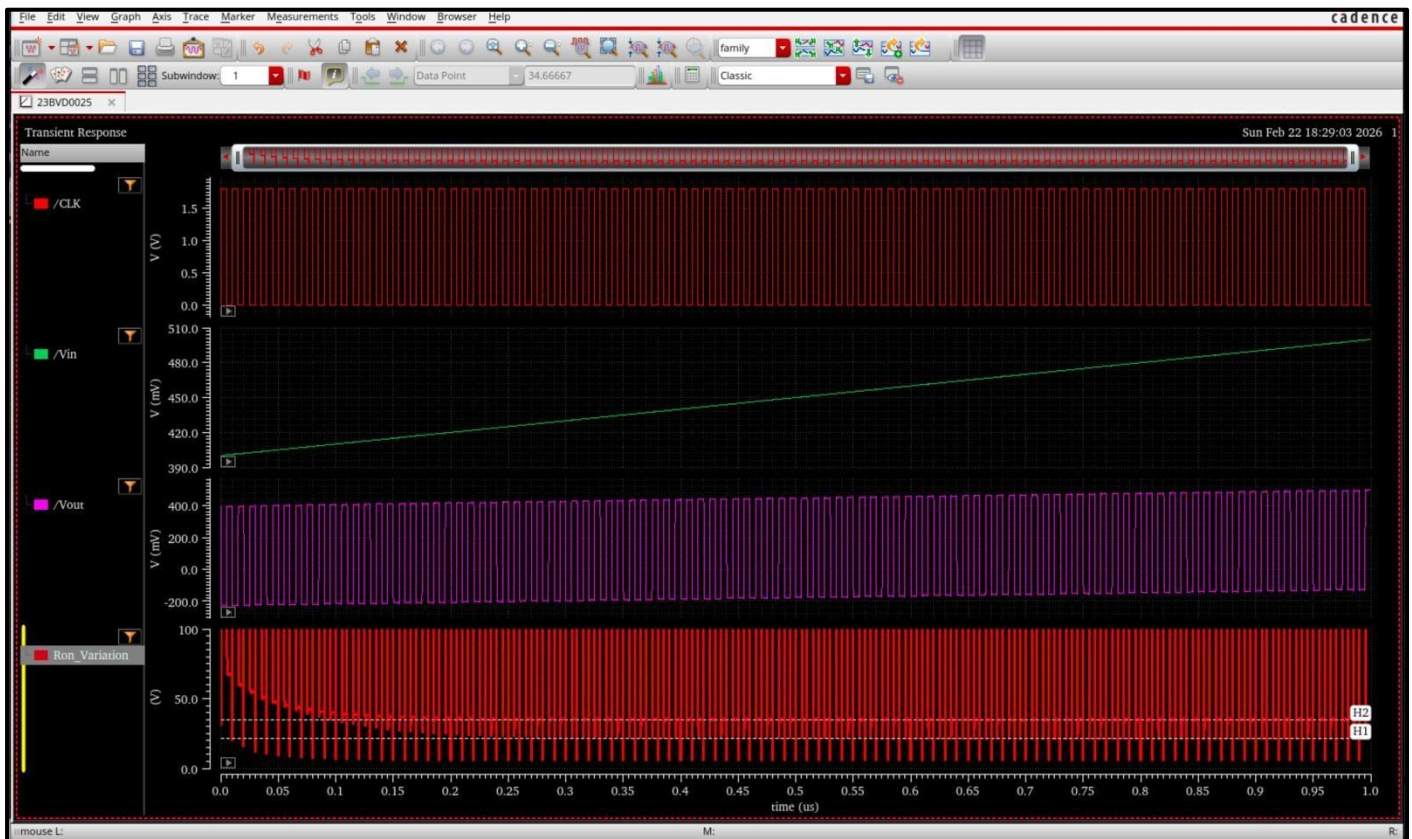
Average Resistance:

$$R_{avg} = \left(\frac{31.96 + 29.84}{2} \right) = 30.9 \, \Omega$$

Percentage Variation:

$$\% variation = \left(\frac{31.86 - 28.54}{30.9} \right) \times 100 = 6.86\%$$

So, variation is $\pm 3.43\%$



Conclusion (III)

The on-resistance variation is less than $\pm 5\%$ across the full input voltage range, satisfying the design requirement and confirming the effectiveness of the bootstrapped architecture.

Conclusion:

Specification	Required	Achieved
R_{ON} Variation	$\leq \pm 5\%$	$\pm 3.43\%$
THD @ 10 MHz	≤ -55 dB	-75.996 dB
Input Frequency	≥ 10 MHz	Pass

A bootstrapped NMOS sampling switch was successfully designed and simulated using a 180 nm CMOS process. The circuit demonstrates correct sample-and-hold operation, near-constant on-resistance, and low harmonic distortion at high input frequencies. These results confirm that the proposed design is well-suited for use in high-speed ADC front-end applications.